INCH POUND
MIL-PRF-3098/7J
05 March 2010
SUPERSEDING
MIL-PRF-3098/7H
19 December 2003

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR25/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 200 kHz to 555 kHz; fundamental: noncontrolled; series resonance.

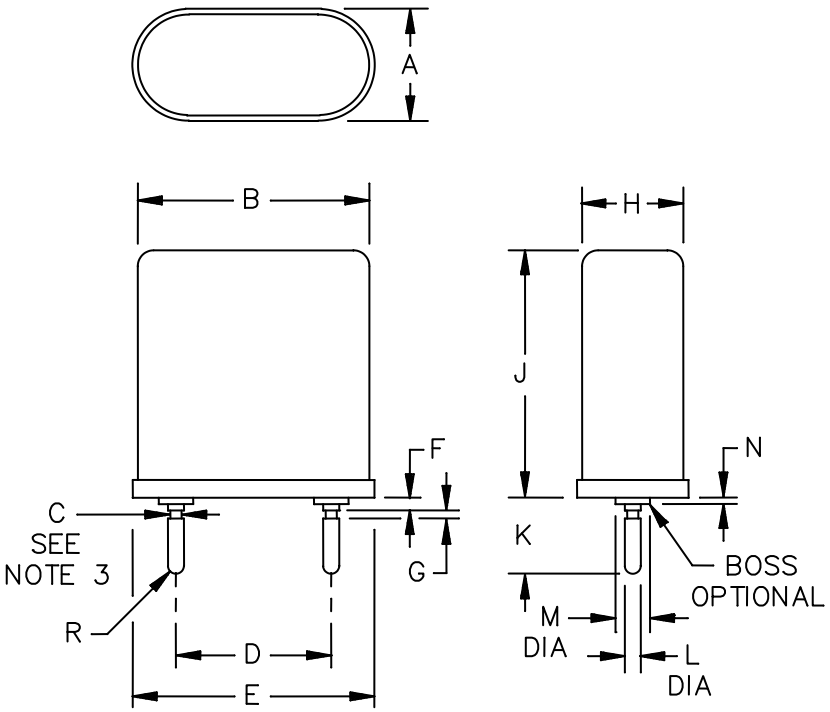


FIGURE 1. Crystal unit - CR25/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.352	---	8.94
B	---	.725	---	18.42
C	.030	.037	0.76	0.94
D	.478	.494	12.14	12.55
E	---	.757	---	19.23
F	.030	.040	0.76	1.02
G	.015	.025	0.38	0.64
H	---	.317	---	8.05
J	---	.775	---	19.69
K	.223	.248	5.66	6.30
L	.048	.052	1.22	1.32
M	.075	.141	1.91	3.58
N	.015	.025	0.38	0.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR25/U - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 200 kHz to 555 kHz, inclusive.

Frequency tolerance: ±100 parts per million (ppm).

Equivalent resistance: See table II.

Mode of oscillation: Fundamental.

Operating temperature range (noncontrolled): -40°C to +85°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Shock (specified pulse):

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Temperature run:

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Bond strength: See table I.

TABLE I. Bond strength.

Frequency range, inclusive (kHz)	Grams, minimum
200 to 250	700
250+ to 320	500
320+ to 370	400
370+ to 435	300
435+ to 555	250

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TABLE II. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
kHz	Ohms
200 to 225	2,500
225+ to 265	3,000
265+ to 290	3,500
290+ to 330	4,000
330+ to 370	4,500
370+ to 410	5,000
410+ to 425	5,500
425+ to 460	6,500
460+ to 555	7,500

Part or Identifying Number (PIN): CR25/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Referenced documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue..

Custodians:
Army - CR
Navy - EC
Air force - 99
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil/

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